

COMMONWEALTH GAMES 2026: TECHNICAL DOCUMENTATION & USER GUIDE

Document Control

- **Project Name:** India's CWG 2026 Performance Analysis
- **Document Type:** Data Methodology & User Operating Manual
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- **Primary Tool:** Microsoft Excel (Pivot Tables, Data Modeling & Accessibility Optimization)

1. Data Cleaning & Transformation Pipeline

To ensure data integrity prior to analysis, the raw dataset underwent a structured cleaning process:

- **Text Standardization:** Normalized discipline and event names across all records using TRIM and case-formatting to remove trailing spaces and inconsistencies.
- **Categorization:** Separated conventional sports disciplines from Para-sport events to enable distinct aggregation.
- **Handling Null Values:** Cleaned missing or unassigned athlete values to prevent calculation skew in summary reporting.

2. Backend Architecture & Pivot Logic

Unlike traditional lookup-heavy workbooks, this project utilizes a **purely Pivot Table-driven backend architecture** for high computational efficiency and seamless performance:

- **Pivot Data Modeling:** All macro-level KPIs (Total Medals, Gold/Silver/Bronze breakdown, Gender distributions) are dynamically summarized directly from the underlying clean dataset via structured Pivot Tables.
- **Accessibility & Metadata Integration (Alt Text):** To maintain professional design standards and accessibility compliance, every key Pivot Table and Visual Chart includes dedicated **Alt Text Descriptions**. This ensures that screen readers and technical reviewers can instantly understand the data model context without breaking visual layouts.

3. Dashboard User Operating Guide

Follow these steps to navigate and interact with the Excel Dashboard in

01_EXCEL_DASHBOARD:

1. **Interactive Filtering (Slicers):**
 - **Gender Slicer:** Click on *Men* or *Women* to isolate performance metrics by gender category.
 - **Discipline Slicer:** Select a specific sport (e.g., *Boxing*, *Athletics*) to inspect discipline-specific medal yields.

2. Reading KPI Cards:

- The top metric cards update dynamically based on your Slicer selection, displaying total medal volume, gold count, and efficiency metrics.

3. Resetting Filters:

- To restore the default view showing all 39 medals, click the **Clear Filter** icon (funnel with a red cross) located at the top-right corner of any Slicer.

4. Data Verification & Integrity

- **Validation Match:** The backend Pivot summaries and front-end Dashboard KPI indicators reconcile 100% against the raw master dataset (Total: 39 Medals).
- **File Optimization:** The workbook uses native Excel Pivot caching to keep file size minimal and ensure fast loading times.